ISM_NSP4_Lecture16_29.08.2026

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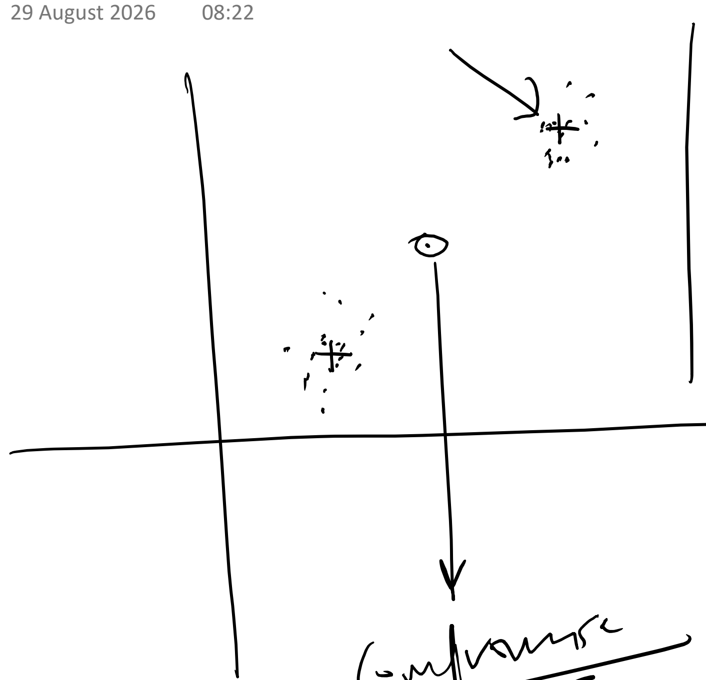
lot of data and you need to guess how the data was generated

↪ out of a distribution

↓

what are the parameters ? this distribution (μ, Σ)



this data came out of $\boxed{\text{a single Gaussian}}$

$$\frac{1}{2\pi|\Sigma|^{\frac{1}{2}}} e^{-(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

Compromise solution

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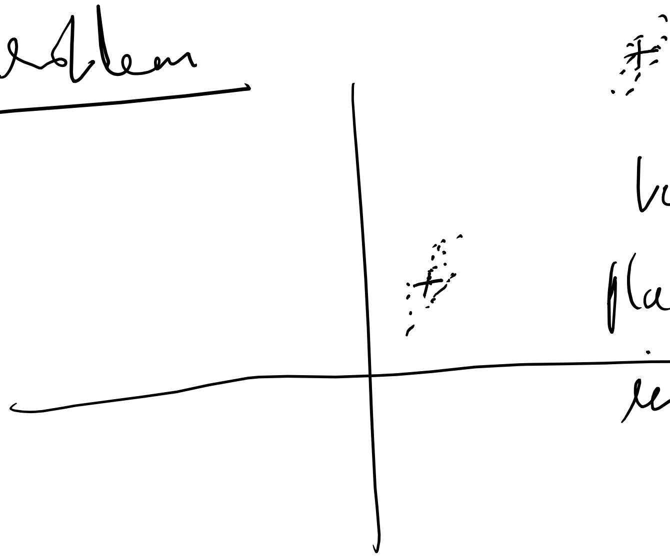
Let us now use two Gaussians for the same problem



best solution is to place each Gaussian in the center of its own cluster.

We assume that the number of Gaussians is fixed [somehow it is known beforehand].

- Given a fixed number of Gaussians, find the parameters $[\pi_k, \mu_k, \Sigma_k]$, $k = 1$ to K

# of Gaussians

π_k is a probability vector denoting how to choose each Gaussian

$$\begin{bmatrix} 0.25, & 0.5, & 0.25 \end{bmatrix} \longleftarrow \pi_k$$

Sample the above vector to identify which Gaussian to use.

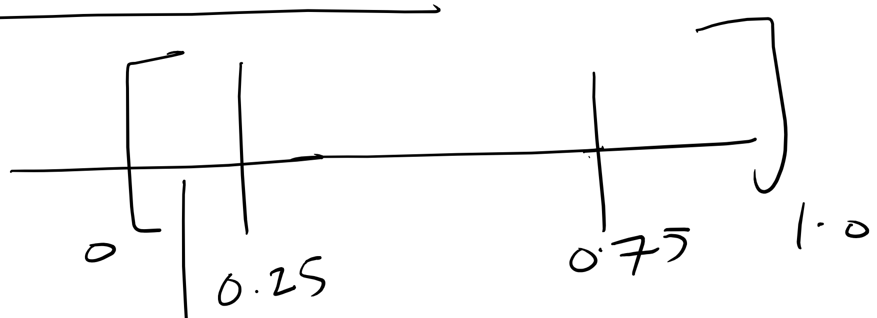
Once you get a Gaussian, sample the Gaussian to get a point $N(\mu_i, \Sigma_i)$

repeat

① first identify the Gaussian

② Then sample the Gaussian

To identify the Gaussian roll a K-sided non-uniform die



random number between 0 and 1

drand48() in C

$0 \le x < 0.25$ } choose Gaussian 1

$0.25 \le x < 0.75$ } choose Gaussian 2

$0.75 \le x < 1.0$ } choose Gaussian 3

Question → How do you know that the sequence of E step → M step → E step → M step will actually improve the likelihood of generating the given data?

$$p(x) = \sum p(x, z)$$

$z \to$ which component to be used

$$\sum_{z=1}^{K} p(x, z) = \sum \underbrace{p(z)}\, p(x/z)$$

$$\pi_K \, p(x/z)$$

$$\boxed{p(x/z=k) = N(\mu_K, \Sigma_K)}$$

$$P(X = x) = P(X = x, Z = 1) + P(X = x, Z = 2)$$
$$\underbrace{+ \cdots P(X = x, Z = k)}$$

these are all mutually exclusive events

Third Axiom being used here

$$P(X = x, Z = 1) = P(Z = 1)\, P(X = x / Z = 1)$$

$$\pi_1 \, P(X = x / Z = 1)$$

$$\downarrow$$

$$N(\mu_1, \Sigma_1)$$

$$\int p(x)\,dx = \int_{-\infty}^{\infty} \sum \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)\,dx$$

$$\underbrace{\pi_1 \int_{-\infty}^{\infty} \mathcal{N}(x \mid \mu_1, \Sigma_1)\,dx}_{=1} + \underbrace{\pi_2 \int_{-\infty}^{\infty} \mathcal{N}(x \mid \mu_2, \Sigma_2)\,dx + \cdots}_{=1}$$

$$= \underline{\pi_1 + \pi_2 + \cdots + \pi_K = 1}$$

Likelihood function $l(\theta)$ is the probability of generating the entire data set

$$P(D|\underline{\theta}) = P(x_1, x_2, \ldots x_N|\theta)$$

$$= \prod_{i=1}^{N} P(x_i|\theta)$$ if the X_is were generated independently

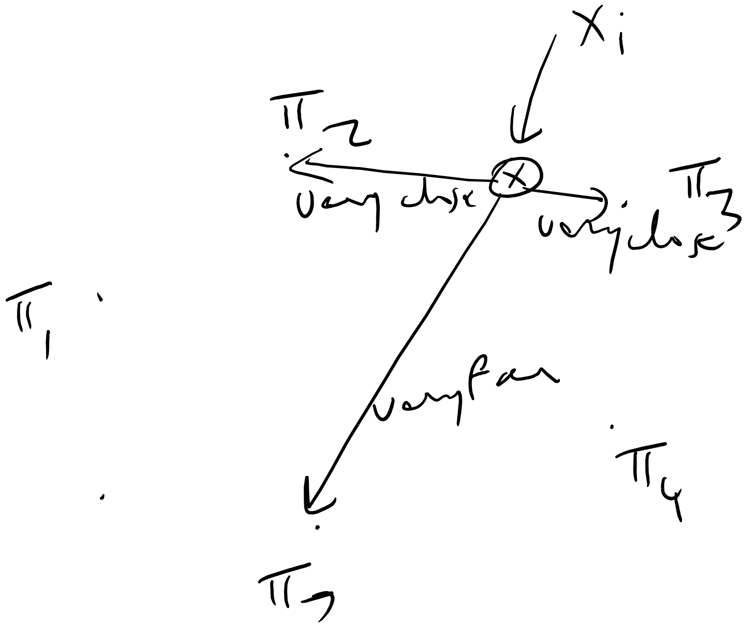
If there is only one Gaussian which generated all of the data, the best answer is

$$\mu = \frac{\sum X_i}{N} = \text{centre of mass of all the data points}$$

$$\boxed{\Sigma = \frac{\sum_{i=1}^{N} (X_i - \mu)(X_i - \mu)^T}{N}}$$

Note that $(X_i - \mu)$ is a column vector

$(X_i - \mu) \to \begin{bmatrix} \\ \\ \end{bmatrix}_{D \times 1}$ $(X_i - \mu)^T \to 1 \times D$



prior probs of choosing the Gaussians

$P(Z_2 = 1 / x_i)$ and $P(Z_3 = 1 / x_i)$ will be large values

$P(Z_5 = 1 / x_i)$ will be a small value

$$\ln\left[P\left(X_1, X_2, \dots X_N / \pi, \mu, \Sigma\right)\right]$$

$\ln ab$
$= \ln a + \ln b$

$$= \ln\left[\prod_{n=1}^{N} P\left(X_n / \pi, \mu, \Sigma\right)\right]$$

$$\sum_{n=1}^{N} \ln\left[P\left(X_n / \pi, \mu, \Sigma\right)\right]$$

But $P\left(X_n / \pi, \mu, \Sigma\right) = \sum_{k=1}^{K} \pi_k N\left(X_n / \pi, \mu, \Sigma\right)$

Singularities in the likelihood function

→ What happens if a μ_j is chosen to be exactly one of the data points, and $\sigma_j \to 0$

Then $\underbrace{\frac{1}{\sigma_j\sqrt{2\pi}}}_{\to\infty} \underbrace{e^{-\frac{1}{2}\left(\frac{x_j-\mu_j}{\sigma_j}\right)^2}}_{=1}$

Singularity

Bayesian treatment assumes $p(\theta)$ over the parameter vector

$\int p(x|\theta)p(\theta)d\theta$ } optimizes

ML → maximum likelihood looks for a single θ that optimizes $\ell(\theta)$ and it may happen that

$\max_\theta \ell(\theta) = \infty$.

$$\mu_K = \frac{1}{N_K} \sum_{n=1}^{N} \gamma(z_{nK}) x_n$$

weighted sum of all data points where the weighting coefficients are individual responsibilities

$$N_K = \sum_{n=1}^{N} \gamma(z_{nK})$$

If we had only one Gaussian

$\gamma(z_{nK}) = 1$ for all n

Then $\underline{N_K = N}$, and

$$\mu_K = \frac{1}{N_K} \sum_{n=1}^{N} \gamma(z_{nK}) x_n$$

$$= \boxed{\frac{1}{N} \sum_{n=1}^{N} x_n}$$

centre of mass of all points

how do we know that the E-step, M-step mechanism actually improves things?

E-step evaluates the log likelihood for the current set of parameters

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$$\ell(\theta) = \log P(X|\theta) = \log \sum_{z} P(X, z|\theta)$$

$$= \log \sum_{z} q(z) \frac{P(X, z|\theta)}{q(z)}$$

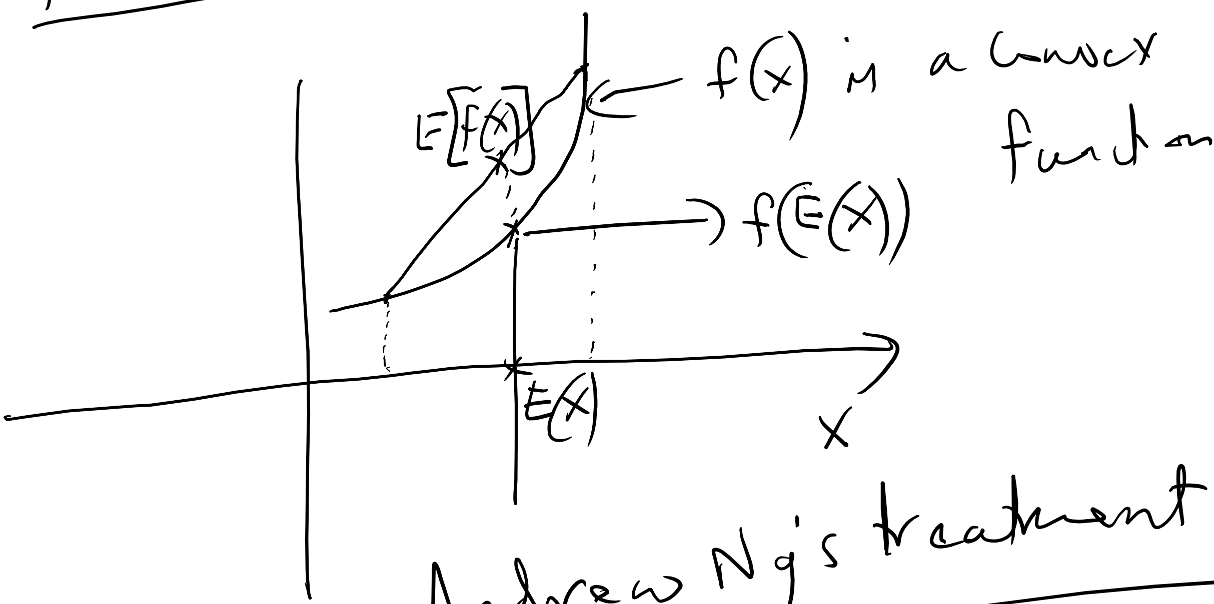
You may have heard of Jensen's inequality → what is it?

Suppose f is a convex function

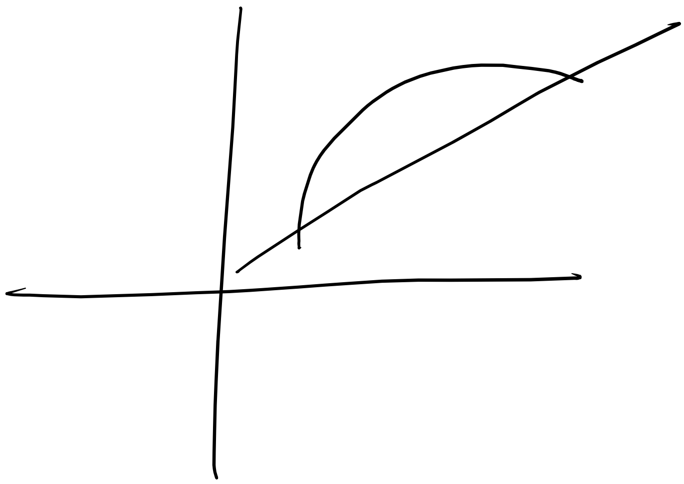
Jensen's inequality states

$$f(E(X)) \leq E[f(X)]$$

X is a random variable



look at Andrew Ng's treatment of this!



In a concave function

$$f(E(x)) \geq E[f(x)]$$

Now log x is a concave fn

Second derivative of log x is -ve

Hessian of log x is -ve

$$\frac{d}{dx} \log x = \frac{1}{x}$$

$$\frac{d^2}{dx^2} \log x = \frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2} < 0$$

$$\log \sum_z q(z) \frac{p(X, z/\theta)}{q(z)}$$

$$E\left[\frac{p(X, z/\theta)}{q(z)}\right]$$

$f[E(Y)]$ is a log fn → concave

$$\underbrace{l(\theta)}_{f(E(X))} \geq \underbrace{\sum_z q(z) \log \frac{p(X, z/\theta)}{q(z)}}_{E[f(X)]}$$ Jensen's inequality

we call the right hand side a function

$$F(\underline{q}, \underline{\theta}) = \sum_z q(z) \frac{p(X, z/\theta)}{q(z)}$$

We can show that in general $l(\theta)$ $\geq F(q, \theta)$ [by Jensen's inequality]

but when $q(z) = p(z/X, \theta)$ then

but when $q(z) = p(z/X, \theta)$ then

$$\underline{\ell(\theta) = F(q, \theta)}$$

What are the E and M steps doing

(1) The E-step chooses

$$q^t(z) = P(Z/X, \theta^t) \text{ because}$$

this makes $\underline{\ell(\theta^t)} = F(q^t, \theta^t)$

(2) The M-step then finds a new parameter value θ^{t+1} satisfying

$$F(q^t, \theta^{t+1}) \geq F(q^t, \theta^t)$$

By Jensen's inequality

$$\ell(\theta^{t+1}) \geq F(q^t, \theta^{t+1})$$

$$\underbrace{\ell(\theta^{t+1}) \geq F(q^t, \theta^{t+1}) \geq \underbrace{F(q^t, \theta^t) = \ell(\theta^t)}_{\text{E-step}}}_{\text{M-step}}$$

E-step: make the lower bound touch the likelihood (by setting $q(z) = p(z/x, \theta)$)

M-step: maximizes the lower bound

$\downarrow$

likelihood cannot decrease